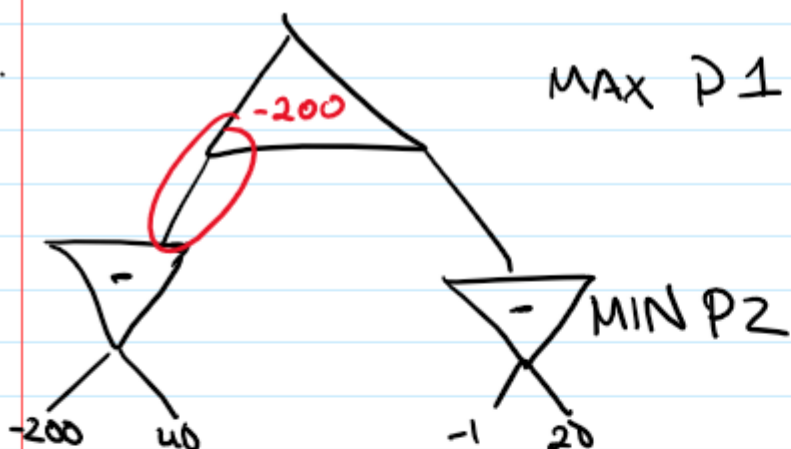
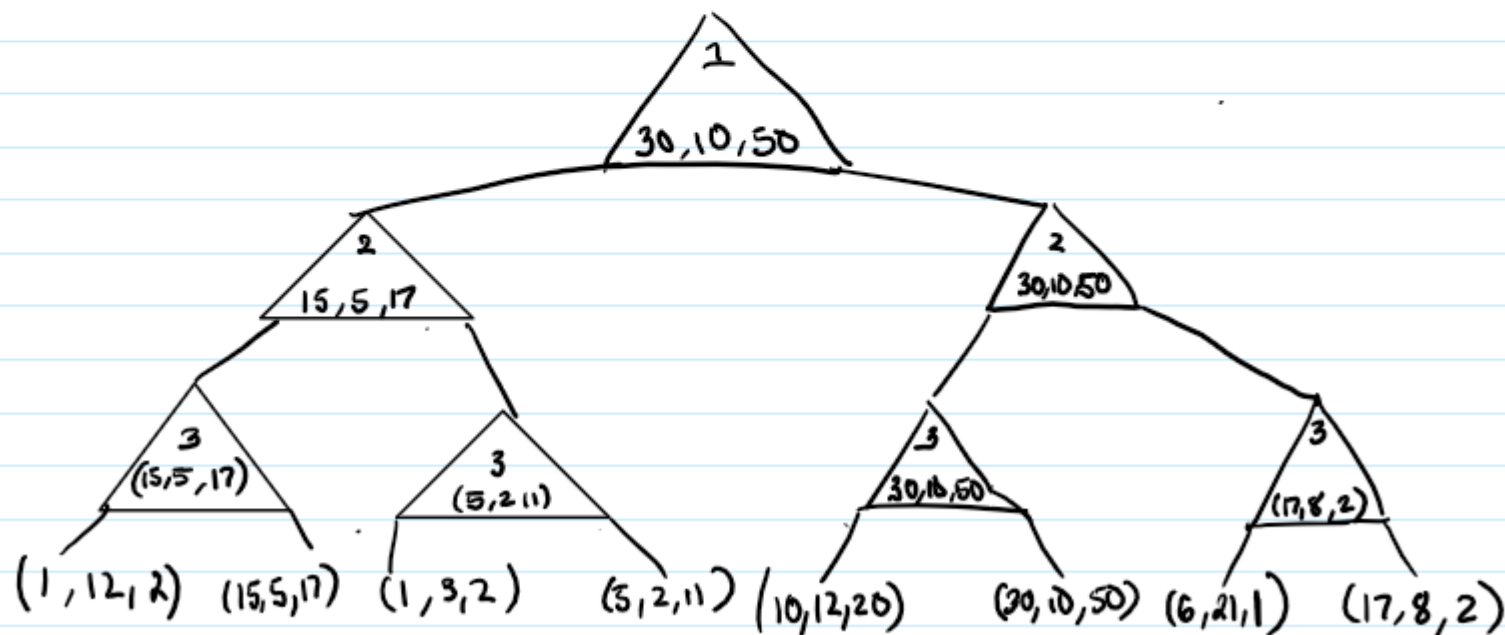


1.



When only considering the sign we see that the minimax algorithm has situations where it goes down the unoptimal path in this case -200 even though the other path is just -1

2.



Since with 3 players we don't have a zero sum game, we can instead represent the utility for each player in a tuple where each player tries to maximize their utility in the step i.e. $U^* = (P1, P2, P3)$

$$\begin{aligned}
 1. \quad P(X_1, X_3 | X_2) &= \frac{P(X_1, X_2, X_3)}{P(X_2)} \quad \text{by definition of conditional prob.} \\
 &= \frac{\cancel{P(X_2)} P(X_1 | X_2) P(X_3 | X_2)}{\cancel{P(X_2)}} \\
 &= P(X_1 | X_2) P(X_3 | X_2) \quad \text{as wanted}
 \end{aligned}$$

$$\begin{aligned}
 2. \quad P(X_1, X_3) &= \sum_{x_2} P(X_1, X_2, X_3) \quad \text{through marginalizing} \\
 &= \sum_{x_2} P(X_2 | X_1, X_3) P(X_1) P(X_3) \\
 &= P(X_1) P(X_3) \sum_{x_2} \cancel{P(X_2 | X_1, X_3)} = 1 \\
 &= P(X_1) P(X_3) \quad \text{as wanted}
 \end{aligned}$$

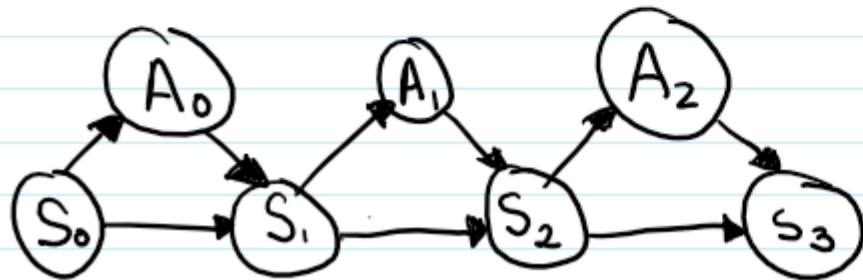
$$\begin{aligned}
 3. \quad \underset{x_1, x_2, x_3}{\operatorname{argmax}} P(X_1, X_2, X_3) &= \underset{x_1, x_2, x_3}{\operatorname{argmax}} P(X_1) P(X_2 | X_1) P(X_3 | X_2) \\
 &= \underset{x_2, x_3}{\operatorname{argmax}} P(X_3 | X_2) \underset{x_1}{\operatorname{argmax}} P(X_1) P(X_2 | X_1) \\
 &= \underset{x_3}{\operatorname{argmax}} \underset{x_2}{\operatorname{argmax}} P(X_3 | X_2) f_1(x_2), \quad f_1(x_2) = \underset{x_1}{\operatorname{argmax}} P(X_1) P(X_2 | X_1) \\
 &= \underset{x_3}{\operatorname{argmax}} f_2(x_3), \quad f_2(x_3) = \underset{x_2}{\operatorname{argmax}} P(X_3 | X_2) f_1(x_2)
 \end{aligned}$$

$$1. \|T^\pi u - T^\pi v\|_\infty = \|r + \gamma P^\pi u - r + \gamma P^\pi v\|_\infty$$

$$= \|\gamma P^\pi u - \gamma P^\pi v\|_\infty$$

$$= \gamma \|P^\pi(u-v)\|_\infty$$

$$\leq \gamma \|u-v\|_\infty, \text{ since } P^\pi \text{ is a transition and therefore each row is a convex combination of } u-v \text{ and so } \|P^\pi(u-v)\|_\infty \leq \|u-v\|_\infty$$



$$3. \nabla_{\pi} J(\pi) = \nabla_{\pi} \sum_{\tau} [G(\tau) P^{\pi}(\tau)]$$

$$= \sum_{\tau} [G(\tau) \nabla_{\pi} P^{\pi}(\tau)] \quad , \quad \nabla \log P^{\pi}(\tau) = \frac{1}{P^{\pi}(\tau)} \nabla P^{\pi}(\tau) \Rightarrow P^{\pi}(\tau) \nabla_{\pi} \log P^{\pi}(\tau) = \nabla_{\pi} P^{\pi}(\tau)$$

$$= \sum_{\tau} [G(\tau) P^{\pi}(\tau) \nabla \log P^{\pi}(\tau)] \rightarrow \nabla_{\pi} \log P^{\pi}(\tau) = \nabla_{\pi} \log \left[\mu(s_0) \prod_{t=0}^{T-1} \pi(A_t | S_t) \prod_{t=0}^{T-1} P(S_{t+1} | S_t, A_t) \right] \text{, from the } P^{\pi}(\tau) \text{ given in 2.}$$

$$= E [G(\tau) \nabla \log P^{\pi}(\tau)]$$

$$= E \left[G(\tau) \sum_{t=0}^{T-1} \nabla_{\pi} \log [\pi(A_t | S_t)] \right]$$

$$= \nabla_{\pi} \left[\log(\mu(s_0)) + \sum_{t=0}^{T-1} \log[\pi(A_t | S_t)] + \sum_{t=0}^{T-1} \log[P(S_{t+1} | S_t, A_t)] \right] \text{, constant wr.t. } \pi$$

$$= \sum_{t=0}^{T-1} \nabla_{\pi} \log [\pi(A_t | S_t)]$$

$$= E \left[\sum_{t=0}^{T-1} \sum_{k=0}^{T-1} \gamma_r^k(S_k, A_k) \nabla_{\pi} \log [\pi(A_t | S_t)] \right]$$

$$= E \left[\sum_{t=0}^{T-1} \nabla_{\pi} \log [\pi(A_t | S_t)] \left(\sum_{k=0}^{t-1} \gamma_r^k(S_k, A_k) + \sum_{k=t}^{T-1} \gamma_r^k(S_k, A_k) \right) \right]$$

$$= \sum_{t=0}^{T-1} \left[E [G_{0:t-1} \nabla_{\pi} \log [\pi(A_t | S_t)]] + E [G_{t:T-1}(\tau) \nabla \log [\pi(A_t | S_t)]] \right]$$

$$= \sum_{t=0}^{T-1} \left[E [G_{0:t-1}] E [\nabla_{\pi} \log [\pi(a_t | s_t)]] + E [G_{t:T-1}(\tau) \nabla \log [\pi(a_t | s_t)]] \right] \quad a_t \perp R_{0:t-1} | S_t \text{ by Markov Property}$$

$$\hookrightarrow E [\nabla_{\pi} \log [\pi(a_t | s_t)]] = \sum_s P(s_t) E_a [\nabla_{\pi} \log \pi(a_t | s_t) | S_t = s]$$

$$= E_s \left[E_a [\nabla_{\pi} \log(a_t | s_t) | S_t = s] \right]$$

$$= E \left[\sum_a \pi(a | s_t) \frac{\nabla_{\pi} \pi(a | s_t)}{\pi(a | s_t)} \right]$$

$$= E \left[\nabla_{\pi} \sum_a \pi(a | s_t) \right] \Rightarrow \sum_{t=0}^{T-1} E [G_{t:T-1}(\tau) \nabla_{\pi} \log [\pi(a_t | s_t)]]$$

$$= E [\nabla_{\pi} (1)] = E(0) = 0$$

$$= E \left(\sum_{t=0}^{T-1} G_{t:T-1}(\tau) \nabla_{\pi} \log [\pi(a_t | s_t)] \right) \text{ as wanted}$$

$$\begin{aligned}
4. \quad \nabla_{\theta} \log \pi_{\theta}(a|s) &= \nabla_{\theta} \log \left[\frac{\exp \theta^T \phi(s, a)}{\sum_{a' \in \mathcal{A}} \exp \theta^T \phi(s, a')} \right] \\
&= \nabla_{\theta} \left[\theta^T \phi(s, a) - \log \sum_{a' \in \mathcal{A}} \exp \theta^T \phi(s, a') \right] \\
&= \phi(s, a) - \sum_{a'} \underbrace{\frac{\exp \theta^T \phi(s, a')}{\sum_{a'' \in \mathcal{A}} \exp \theta^T \phi(s, a'')}}_{\pi_{\theta}(a', s)} \phi(s, a') \\
&= \phi(s, a) - \sum_{a' \in \mathcal{A}} \pi_{\theta}(a', s) \phi(s, a') \quad , \quad \pi_{\theta}(a', s) = \text{by def.}
\end{aligned}$$